# ACS and STEMI

Territories, time goals, first drugs and complications

**Medic Basics**
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## ACS SPECTRUM AND STEMI CRITERIA

| TYPE | TROPONIN | ECG |
|---|---|---|
| **Unstable angina** | Not elevated | Normal or ST-T changes |
| **NSTEMI** | Rise/fall above 99th percentile | No ST elevation |
| **STEMI** | Rises; never wait for it | New ST elevation |

- ST elevation ≥1 mm in 2 contiguous leads; V2-V3: ≥2 mm men 40+, ≥2.5 mm men under 40, ≥1.5 mm women
- ST depression in V1-V3: get V7-V9; ≥0.5 mm elevation = posterior STEMI
- Inferior STEMI: get V4R; ≥0.5 mm elevation = RV involvement

| PATTERN | MEANING |
|---|---|
| **LBBB or paced with concordant ST changes (Sgarbossa)** | STEMI equivalent; new LBBB alone is not |
| **de Winter T waves** | Proximal LAD occlusion: cath lab now |
| **Wellens T waves V2-V3, pain-free** | Critical LAD; cath, no stress test |
| **aVR elevation, diffuse ST depression** | Left main or 3-vessel; no lytic |

## 12-LEAD TERRITORIES

| WALL | LEADS | ARTERY | WATCH FOR |
|---|---|---|---|
| **Septal/anterior** | V1-V4 | LAD | Pump failure, VSD, LV thrombus |
| **Lateral** | I, aVL, V5-V6 | LCx or diagonal | Often with other walls |
| **Inferior** | II, III, aVF | RCA (80%) or LCx | Bradycardia, AV block, RV MI |
| **Right ventricle** | V4R; clues V1 elevation, III > II | Proximal RCA | Hypotension: IV fluid, no nitrates |
| **Posterior** | V7-V9 | LCx or RCA | Missed without posterior leads |




Confirm monitor ST elevation on a 12-lead.

Inferior MI block: nodal, transient, narrow escape; atropine may help. Anterior MI block: infranodal, wide unstable escape; prepare to pace. New BBB alone: pads on, watch.

## TIME GOALS (ACC/AHA)

| STEP | GOAL |
|---|---|
| **12-lead read** | ≤10 min from first contact; repeat if nondiagnostic |
| **Primary PCI** | First contact (door, if walk-in) to device ≤90 min |
| **Transfer for PCI** | First contact to device ≤120 min |
| **PCI over 120 min away, symptoms under 12 h** | Fibrinolytic within 30 min of arrival |
| **After fibrinolytic** | Transfer now; angiography at 2-24 h |
| **NSTE-ACS** | Unstable: immediate cath; high risk: ≤24 h |

**EXAM TIP** Remember 10, 90, 120, 30. EMS: transmit the 12-lead, notify early, bypass to a PCI center.

## INITIAL MEDICATIONS

Typical adult doses for exam study; follow your institution's or agency's protocols. EMTs: aspirin and assisting with the patient's own nitroglycerin.

| DRUG | TYPICAL DOSE | KEY POINT |
|---|---|---|
| **Aspirin** | 162-325 mg chewed; then 75-100 mg daily | Even if on daily aspirin |
| **Ticagrelor** | 180 mg load | Preferred over clopidogrel; dyspnea |
| **Prasugrel** | 60 mg load | Also preferred; PCI only; no stroke/TIA; usually avoid 75+ |
| **Clopidogrel** | 600 mg for PCI; 300 mg with lytic | Lytic at 75+: 75 mg, no load |
| **Heparin (UFH)** | PCI: 70-100 units/kg to ACT; lytic: 60 units/kg (max 4,000) | Or bivalirudin, enoxaparin |
| **Nitroglycerin** | 0.4 mg SL every 5 min, max 3 | Only if SBP ≥90 |
| **Oxygen** | Only if SpO2 under 90% | No routine use |
| **Beta-blocker** | Oral, within 24 h | Not with HF, shock risk or block |

**SAFETY** No nitrates: SBP under 90 (or 30 below baseline), RV infarct, sildenafil or vardenafil in past 24 h, tadalafil 48 h, avanafil 12 h. Opioids only if nitrates fail. No NSAIDs except aspirin.

## FIBRINOLYSIS AND REPERFUSION

Absolute contraindications:

- Any prior intracranial hemorrhage
- Cerebral AVM or malignant brain tumor
- Ischemic stroke within 3 months
- Suspected aortic dissection
- Active bleeding or bleeding disorder (not menses)
- Head/face trauma within 3 months; brain/spine surgery within 2 months
- BP over 180/110 despite emergency treatment

Relative: BP over 180/110 on arrival, oral anticoagulant, recent internal bleeding, pregnancy, prolonged CPR. No lytic for ST depression alone unless posterior STEMI. Reperfusion signs at 60-90 min:

- Pain relief
- ST elevation down ≥50% (2025 ACC/AHA: ≥70% if inferior)
- Reperfusion rhythm: classically AIVR, usually benign

**EXAM TIP** Failed reperfusion (ongoing pain, poor ST resolution, instability): immediate rescue PCI.

**SAFETY** Post-lytic headache, confusion or new deficit: suspect a brain bleed; stop antithrombotics, emergency head CT.

## COMPLICATIONS

| COMPLICATION | TIMING | CLUES |
|---|---|---|
| **VF or pulseless VT** | Most within 48 h | Defibrillate per ACLS |
| **Papillary muscle rupture** | 3-5 days | Inferior/lateral MI; pulmonary edema; murmur may be absent |
| **Septal rupture (VSD)** | 3-5 days | New pansystolic murmur; O2 step-up RA to PA |
| **Free wall rupture** | 3-5 days; 24-48 h after lytics | Tamponade, PEA; late reperfusion raises risk |
| **Cardiogenic shock** | Any time | 5-10% of acute MI; 40-50% short-term death |

- Shock: emergency culprit PCI or CABG, any time; no routine non-culprit PCI
- Impella: reasonable in selected STEMI shock; more bleeding, limb ischemia and renal replacement (such as CRRT)
- Routine IABP or VA-ECMO: no survival benefit

**SAFETY** New murmur, flash pulmonary edema or collapse days after MI: urgent echo, surgical center, MCS as a bridge.

**Sources** (23) are listed with links at medicbasics.com/sheets/acs-stemi.html. For education and exam preparation only. Not medical advice; follow current guidelines and your institution's protocols. Medic Basics is an independent study aid not affiliated with any certifying body.